

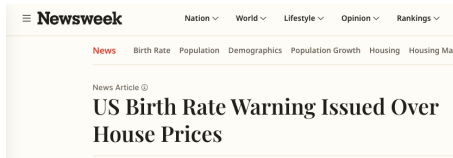
# Fertility and Housing Tenure Choice

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Macroeconomics Student Lunch

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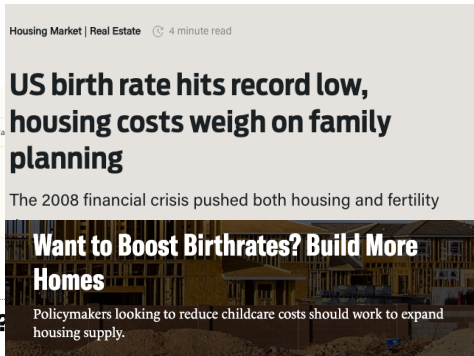
# Housing and Fertility



POLICY

## Will cheap housing lead to more babies?

The connection between housing and fertility rates has a missing piece.



# Renting a home 'leads to lower birth rates'

# Housing, Fertility, and Tenure Choice

**Housing costs affect fertility**—but tenure and wealth matter

- Owners have more kids when house prices or housing wealth rise; renters do not (Dettling & Kearney 2014), and big units matter (Couillard 2025)

Children increase how big of a house you need. This has implications for:

- Owning: tenure segmentation
- Location: housing cost differs geographically
- Wealth: borrowing constraints, bequests

Physical space is key. How much does it affect fertility?

What kind of housing policy is good for natality?

→ focus on space constraints and tenure choice

# This Paper

Need a structural model where tenure, location, and fertility are joint. Today:

**Model:** lifecycle model of fertility, location, tenure, and saving

- Childbirth is a discrete shock to housing need
- Housing markets with borrowing constraints, tenure choice, transaction costs
- Location and tenure respond to family-space need

**Empirics:** document facts about fertility, housing, and moving

- ACS: parents less central; re-sort toward periphery within metros
- PSID: first birth raises ownership, rooms, and moves for space

**Goal:** write a model disciplining these interactions, think about policy.

**Model**

# Environment

- Two locations: **center** ( $C$ ) and **periphery** ( $P$ ) within a metro area
  - Center has an amenity premium but less elastic housing supply
  - Common wages; sorting driven by amenities, housing costs, and moving frictions
- Adult age  $a \in \{0, 1, \dots, J\}$ ; deterministic aging and death
  - Working age  $a \in [0, J_R)$ ; fertile window  $a \in [J_f^{\text{start}}, J_f^{\text{end}}]$
  - Children do not work; they depend on their parents
- Households choose:
  - **Fertility**:  $n^* \in \{0, 1, \dots, \bar{n}\}$ , subject to EV shocks
  - **Location**: center or periphery each period (with moving costs and EV shocks)
  - **Tenure**: rent or buy a discrete house size  $H_k$  on a housing ladder
  - **Saving**: liquid wealth  $b$  with collateral borrowing for owners
- Children age stochastically through life stages; housing need depends on children
- Benchmark city scale is normalized to 1; counterfactual city size responds to entry from city-born and outside-origin young adults

# Preferences

$n^*$ : completed fertility (chosen once);  $s$ : child-aging stage (stochastic);  $m = m(n^*, s)$ : children currently at home.

**Flow utility.** Over consumption and housing services:

$$u(c, \mathbf{h}; m) = \frac{\left[ (c - \bar{c}(m))^\alpha (\mathbf{h} - \bar{h}(m))^{1-\alpha} \right]^{1-\sigma}}{1 - \sigma} + \psi_{\text{child}} m.$$

**Subsistence requirements.** Depend on family size:

$$\bar{c}(m) = \bar{c}_0 + \bar{c}_1 m, \quad \bar{h}(m) = \bar{h}_0 + \underbrace{\bar{h}_{\text{jump}} \mathbf{1}\{m \geq 1\}}_{\text{first-child housing margin}} + \bar{h}_1 m.$$

**Housing services.** Households either rent or own; owners get a service premium:

$$\mathbf{h} = \underbrace{\mathbf{1}\{\text{own}\} \chi h}_{\text{owner services}} + \underbrace{\mathbf{1}\{\text{rent}\} h^R}_{\text{renter services}}, \quad \chi > 1.$$

- Child costs disappear after children mature.

# Bequests

Terminal utility at death:

$$\mathcal{B}(\tilde{b}, n^*) = \theta_0 (1 + \theta_n n^*) \frac{(\theta_1 + \tilde{b})^{1-\sigma}}{1-\sigma}, \quad \tilde{b} = b + (1 - \psi)p_i h.$$

- $\theta_0$  governs overall bequest motive (saving level).
- $\theta_n$  governs the parent–childless saving differential.
- Total bequeathable wealth  $\tilde{b}$  includes housing equity net of the transaction cost for owners.

# Housing, Tenure, and Budget Constraints

Renters choose housing continuously:

$$h^R \in [0, \bar{h}^R].$$

Owners choose a discrete house size:

$$h \in \{H_1, \dots, H_{N_H}\}.$$

## Renters

$$b' = Rb + y(a) - c - r_i h^R, \quad b' \geq 0.$$

## Owners

$$b' = Rb + y(a) - c - (\delta + \tau_H)p_i h' + \underbrace{\mathbf{1}\{\text{sell}\}(1 - \psi)p_i h}_{\text{sale proceeds}} - \underbrace{\mathbf{1}\{\text{buy}\} p_i h'}_{\text{purchase cost}}, \quad b' \geq -\phi p_i h'.$$

- **Tenure segmentation:** renters access housing up to  $\bar{h}^R$ ; owner sizes start at  $H_1 > \bar{h}^R$ .
- Movers and newly matured children arrive as renters, then re-choose tenure and size.

## Fertility and Child Aging

- Fertility choices are made only in the fertile window  $a \in [J_f^{\text{start}}, J_f^{\text{end}}]$ .
- One-shot fertility: can have multiple kids, but choose only once:

$$n^* \in \{0, 1, \dots, \bar{n}\}.$$

- After birth, children age through stages  $s$  (think toddler  $\rightarrow$  child  $\rightarrow$  teen  $\rightarrow$  adult):
  - Aging is **sequential**: each stage transitions only to the next one, with random arrival
  - All  $n^*$  children of a parent age **together** as a single cohort -  $s$  is shared.
- While children are at home ( $s < \text{matured}$ ), they raise housing need  $\bar{h}(m)$  and consumption cost  $\bar{c}(m)$ .
- Once matured, child costs shut off and children enter the economy at age  $a = 0$ .

# Firms and Earnings

- Competitive firms produce with labor:

$$Y = AL, \quad w = A.$$

- Common wages** across locations: sorting driven entirely by amenities and housing costs.
- Age-dependent earnings:

$$y(a) = \begin{cases} (1 - \tau_{\text{pay}}) w(a) & a < J_R \\ \bar{b} & a \geq J_R \end{cases}$$

- Retirees receive a stationary pay-as-you-go pension  $\bar{b}$  financed by payroll taxes.

# Housing Supply and Price Mapping

Housing supply in each location:

$$H_i^s(r_i) = \bar{r}_i r_i^{\eta_i}, \quad i \in \{C, P\}.$$

- Less elastic center supply makes family space more expensive in the core.
- House prices map into user costs through

$$r_i = (R + \delta + \tau_H)p_i.$$

# Entry Environment

- **Potential entrants.** At the beginning of adult life, potential entrants are maturing children raised in the modeled metro and a mass  $M$  of outside-origin potential immigrants.
- **Choice.** Each potential entrant chooses where to start adult life:

$$\{C, P, E\}.$$

Entering location  $i \in \{C, P\}$  gives value  $W_i^E(p)$  and the outside economy gives value  $\bar{W}^E$ .

- **Entry probabilities.** With extreme-value entry tastes,

$$\pi_C^E(p) + \pi_P^E(p) + \pi_{\text{out}}^E(p) = 1, \quad q^E(p) = \pi_C^E(p) + \pi_P^E(p) = 1 - \pi_{\text{out}}^E(p).$$

The benchmark chooses  $\bar{W}^E$  to match the outside margin, then holds it fixed in counterfactuals.

# Entry and City Scale

1. Let  $g(p)$  be the normalized stationary composition and let  $S$  be total adult mass, so the level distribution is  $G(p) = Sg(p)$ .
2. Per unit of city scale, stationarity requires replacement inflow  $E_0(p)$  of young adults; equivalently, this replaces adults who exit.
3. Per unit of city scale, households also produce  $B_0(p)$  mature city-born potential entrants.
4. At city scale  $S$ , the potential entrant pool is  $M + SB_0(p)$ .
5. Since only fraction  $q^E(p)$  chooses the modeled metro, stationarity requires

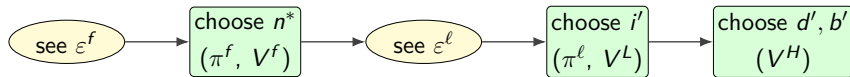
$$SE_0(p) = q^E(p) [M + SB_0(p)].$$

6. Hence counterfactual city scale is

$$S(p) = \frac{q^E(p)M}{E_0(p) - q^E(p)B_0(p)}.$$

**Benchmark.** Set  $S^* = 1$ . The normalized benchmark implies  $E_0^*$  and  $B_0^*$ ; the outside value pins down  $q^{E,*}$ ; then  $M$  is chosen residually. In counterfactuals, hold  $(M, \bar{W}^E, \kappa_E)$  fixed.

# The Household's Problem



The full household state is

$$x = (b, d, i, a, n^*, s).$$

- $b$ : liquid wealth;  $d$ : tenure;  $i$ : location;  $a$ : age.
- $n^*$ : completed fertility;  $s$ : child-aging stage.

Choices nest as: **fertility**  $\rightarrow$  **location**  $\rightarrow$  **housing / saving**.

# The Household's Problem: Housing and Saving



Given  $(b, d, i, a, n^*, s)$ , jointly choose housing and next assets:

$$V^H(b, d, i, a, n^*, s) = \max_{d', b'} \left\{ u(c, \mathbf{h}(d'); m(n^*, s)) + \beta \mathbb{E}_{s'} [V(b', d', i, a + 1, n^*, s')] \right\}$$

Let  $H(R) = 0$  and  $H(O_k) = H_k$ . Feasible branches:

- **Rent:**  $d' = R$ ,  $\mathbf{h}(d') = h^R \in [0, \bar{h}^R]$ ,

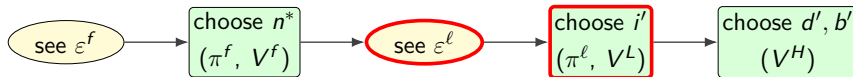
$$c + b' + r_i h^R = Rb + y(a) + \mathbf{1}\{d \neq R\}(1 - \psi)p_i H(d), \quad b' \geq 0.$$

- **Own:**  $d' = O_k$ ,  $\mathbf{h}(d') = \chi H_k$ ,

$$c + b' + (\delta + \tau_H)p_i H_k = Rb + y(a) + \mathbf{1}\{d' \neq d\}[(1 - \psi)p_i H(d) - p_i H_k], \quad b' \geq -\phi p_i H_k.$$

Movers arrive here as renters before choosing  $d', b'$ .

# The Household's Problem: Location Choice



Draw Gumbel shocks; choose center or periphery:

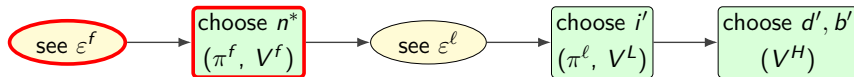
$$V^L(b, d, i, a, n^*, s) = \kappa_\ell \ln \sum_{i' \in \{C, P\}} \exp \left( \frac{V^H(\tilde{b}(i'), \tilde{d}(i'), i', a, n^*, s) + E_{i'}^R - \mu_{ii'}}{\kappa_\ell} \right)$$

- $(\tilde{b}(i'), \tilde{d}(i'))$ : wealth and tenure after choosing destination  $i'$ .
- $E_{i'}^R$ : amenity;  $\mu_{ii'}$ : moving cost;  $\kappa_\ell$ : taste dispersion.

Choice probabilities:

$$\pi_{i'}^\ell(b, d, i, a, n^*, s) = \frac{\exp \left( \frac{V^H(\tilde{b}(i'), \tilde{d}(i'), i', a, n^*, s) + E_{i'}^R - \mu_{ii'}}{\kappa_\ell} \right)}{\sum_{i''} \exp \left( \frac{V^H(\tilde{b}(i''), \tilde{d}(i''), i'', a, n^*, s) + E_{i''}^R - \mu_{ii''}}{\kappa_\ell} \right)}.$$

## The Household's Problem: Fertility Choice



Childless households in the fertile window choose completed fertility once:

$$V(b, d, i, a, 0, 0) = \kappa_n \ln \sum_{n' \in \mathcal{N}} \exp\left(\frac{V^L(b, d, i, a, n', s(n'))}{\kappa_n}\right)$$

Choice probabilities:

$$\pi_{n'}^f(b, d, i, a, 0, 0) = \frac{\exp\left(\frac{V^L(b, d, i, a, n', s(n'))}{\kappa_n}\right)}{\sum_{n''} \exp\left(\frac{V^L(b, d, i, a, n'', s(n''))}{\kappa_n}\right)}.$$

# Sorting and Housing-Demand Elasticity

$\mathcal{E}_C > \mathcal{E}_P$  (amenity);  $r_C > r_P$  (rents);  $M = Rb + y - b'$  (expenditure budget);

$\bar{h}(n)$  housing requirement with  $n$  children,  $\bar{h}(1) > \bar{h}(0)$ . Ignoring moving costs and Gumbel

## Sorting.

- Higher  $\bar{h} \Rightarrow$  larger rent bill in  $C \Rightarrow$  preference for  $C$  falls.
- Single crossing: for each  $M$ , unique threshold  $\hat{h}(M)$  s.t.  $\bar{h} < \hat{h} \Rightarrow C$ ,  $\bar{h} > \hat{h} \Rightarrow P$ .
- $\hat{h}$  rises with  $M$  (richer households absorb high  $r_C$ ).
- Since  $\bar{h}(1) > \bar{h}(0)$ : parents sort weakly peripheral.
  - Parent share falls with local rent.
  - First birth triggers moves to lower- $r$  areas.

**Elasticity.** Housing demand for uncapped renter splits into committed + discretionary:

$$h^*(n) = \underbrace{\alpha \bar{h}(n)}_{\text{committed}} + \underbrace{\frac{(1-\alpha)(M-\bar{c})}{r}}_{\text{discretionary}}.$$

- Parents less income-elastic on intensive margin; less responsive to rent.
- Kids raise the probability of owning.

## Sorting and rents

- In this static derivation, let  $\mathcal{V}_i(n; M)$  be indirect utility for a renter with  $n$  children in location  $i$ .
- $\Delta\mathcal{V}(n; M) \equiv \mathcal{V}_C(n; M) - \mathcal{V}_P(n; M)$
- Sorting wedge:  $\mathcal{W}(M) \equiv \Delta\mathcal{V}(0; M) - \Delta\mathcal{V}(1; M) > 0$
- If the rent gap  $r_C - r_P$  widens, the parent-childless gap in location preferences  $\mathcal{W}$  widens:

$$\left. \frac{\partial \mathcal{W}}{\partial r_C} \right|_{r_P} > 0.$$

- Across cities: the parent-childless center-share gap rises with the local center-periphery rent gap.

# Constrained Squeeze

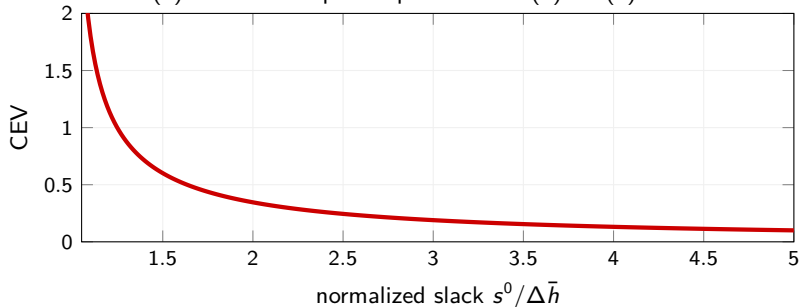
derivation

Capped renters and owners on a fixed rung cannot adjust housing. How costly is a child on a home that barely fits one?

Consumption-equivalent variation, holding housing fixed:

$$u(c + \text{CEV}, h, 1) = u(c, h, 0).$$

Pre-birth slack  $s^0 = h - \bar{h}(0)$  and extra required space  $\Delta \bar{h} = \bar{h}(1) - \bar{h}(0)$ .



# Summary

- A model of fertility, location, tenure, and saving with child-driven housing need shocks
- Children raise housing need, with implications for tenure and location choice
- The sorting wedge between parents and childless rises with the local rent gap.
- The closer a household is to being space-constrained, the more costly a child is
- Some predictions:
  - Parents sort peripheral, are less housing-elastic, and are more likely to own.
  - Sorting strengthens with the local rent gap.

The data can help:

- Improve our understanding of housing and fertility
- Validate the predictions
- Inform our quantification

# Empirics

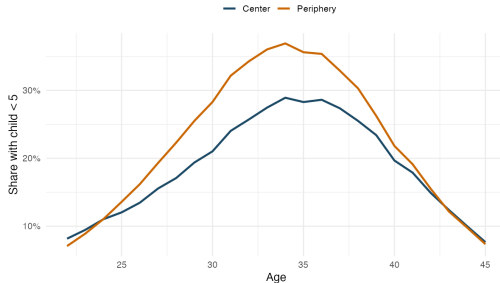
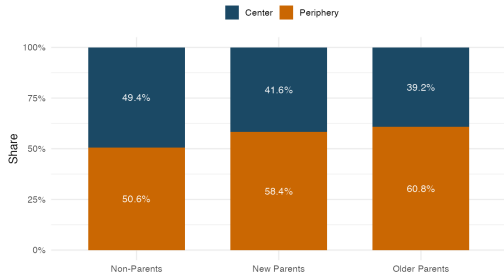
# Data

- **American Community Survey (ACS) 1-year estimates:**
  - Annual survey data collected by the U.S. Census Bureau
  - Geographic detail up to PUMA level
  - Demographic characteristics, socioeconomic indicators, and housing information
- **Panel Study of Income Dynamics (PSID):**
  - Longitudinal household survey initiated in 1968 (24,000 households in 2021)
  - Follows original sample families and their descendants across generations
  - Rich demographics, income and wealth information
  - Just got the geocoded data, so I can track location and moves at the zip level: next time!

# Parenthood and Location

Geography

*Model: parents sort to the periphery*



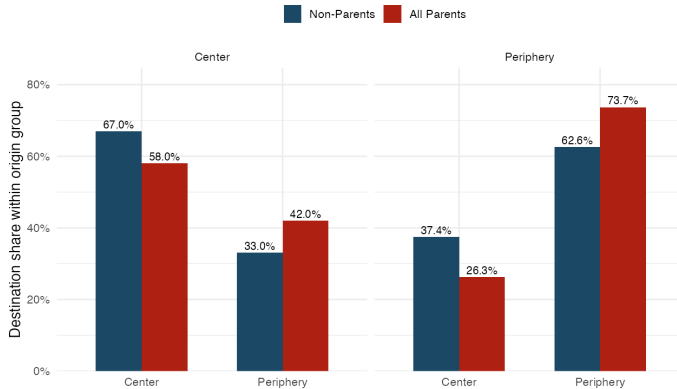
- Non-parents are more concentrated in the center than new parents: 0.494 vs. 0.416.
- More parents outside the center

# From Movers: Parents Re-Sort Within Metro Areas

Regressions

Within moves

*Model: parents sort to the periphery; childbirth triggers moves to lower-r areas.*



- Among within-metro movers, parents are less likely to choose center destinations and more likely to choose peripheral destinations.

Baseline center shares: 49.4% for non-parents, 41.6% for new parents. Owner rates: 49% vs. 59%.

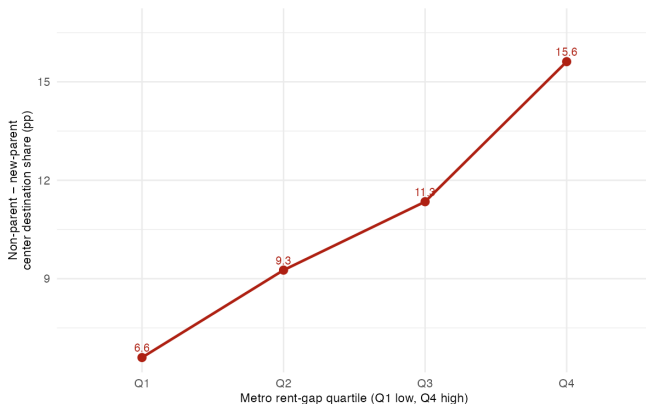
# Sorting Strength and the Local Rent Gap

Regression

Elasticity

Stock

*Model: parent share falls with local rent;  $\partial W / \partial r_C > 0$ .*



Y-axis: non-parent minus new-parent center destination share among within-metro movers (pp). Base destination shares: 52.4% for non-parents, 41.5% for new parents.

- The sorting gap among movers more than doubles from Q1 to Q4.

## Using the PSID: Event-Study Design

For household  $i$ , let  $E_i$  denote the year of childbirth. For any housing outcome  $Y_{it}$ , estimate

$$Y_{it} = \sum_{k=-5, k \neq -2}^{10} \alpha_k \mathbf{1}\{t - E_i = k\} + \zeta_i + \tau_t + \gamma_{a(i,t)} + \delta X_{it} + \varepsilon_{it}.$$

- $\zeta_i$ ,  $\tau_t$ , and  $\gamma_{a(i,t)}$  are household, calendar-year, and age fixed effects.
- The omitted period is  $k = -2$ , so each  $\alpha_k$  is relative to two years before birth.
- I use the same design for homeownership, moving-for-size, and rooms.

# PSID: Childbirth Raises Moves for Space

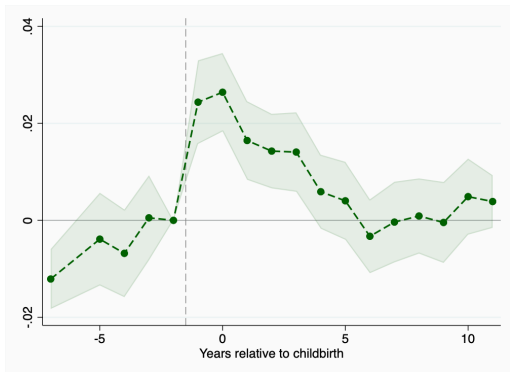
Reasons

By Category

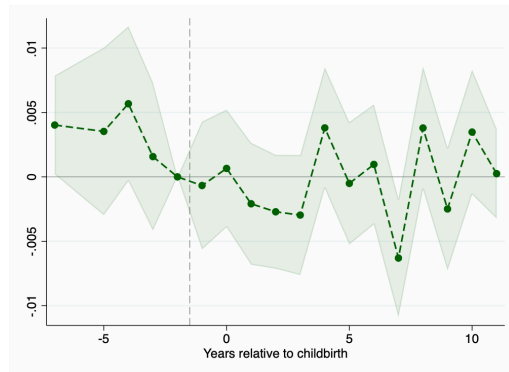
No IDFE

$\hat{\alpha}_k$ : change in probability of moving for reason  $Y$ , relative to  $k = -2$ .

$Y = \mathbf{1}\{\text{moved for expansion of housing}\}$



$Y = \mathbf{1}\{\text{moved for neighborhood quality}\}$



- The probability of moving for housing size rises around first birth.
- By contrast, moves for neighborhood quality do not rise in the same way.

# PSID: Ownership and Housing Size Around First Birth

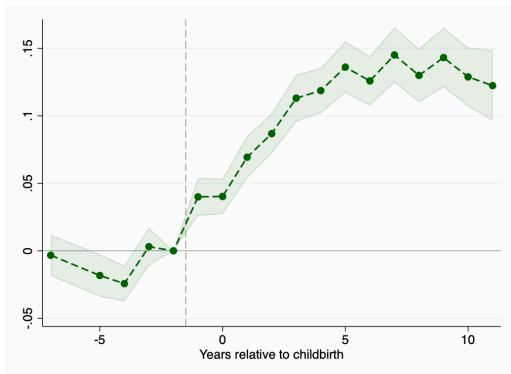
Own: No IDFE

Rooms: No IDFE

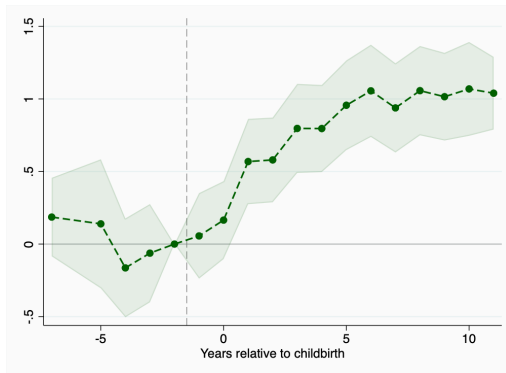
Model:  $\bar{h}(1) > \bar{h}(0)$ ; kids raise ownership.

$\hat{\alpha}_k$ : change relative to  $k = -2$ .

$Y_{it} = \mathbf{1}\{\text{owner}\}_{it}$



$Y_{it} = \text{number of rooms}_{it}$



- Homeownership rises sharply after first birth; rooms expand by about 0.66 at  $k = +3$ .

Pre-birth mean: 6.0 rooms.

# Quantification

## Externally Calibrated Parameters

|                | Param               | Description          | Value    | Source                |
|----------------|---------------------|----------------------|----------|-----------------------|
| <i>Pref.</i>   | $\sigma$            | Risk aversion        | 2.0      | –                     |
|                | $\alpha$            | Consumption weight   | 0.70     | BLS CEX               |
|                | $\bar{c}_0$         | Consumption floor    | 0.10     | Poverty line          |
| <i>Fin.</i>    | $r$                 | Interest rate        | 0.04     | –                     |
|                | $\delta$            | Depreciation         | 0.02     | Greaney et al. (2024) |
|                | $\tau_H$            | Property tax         | 0.01     | US average            |
|                | $\phi$              | LTV limit            | 0.80     | 20% down payment      |
|                | $\psi$              | Transaction cost     | 0.06     | Greaney et al. (2024) |
| <i>Life</i>    | $J, J_R$            | Lifespan, retirement | 60, 47   | Ages 18–77, ret. 65   |
|                | $\tau_{\text{pay}}$ | Payroll tax          | 0.179    | Greaney et al. (2024) |
| <i>Housing</i> | $\eta_P, \eta_C$    | Supply elasticities  | 2.0, 0.8 | Saiz (2010)           |

# Internally Calibrated Parameters

Strategy

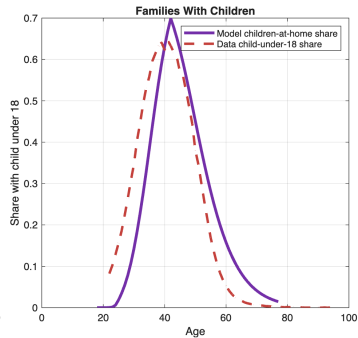
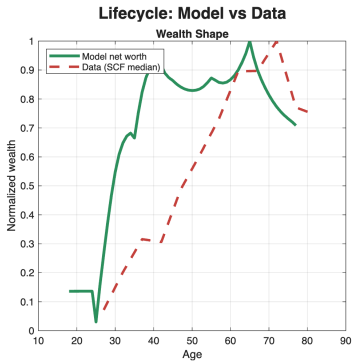
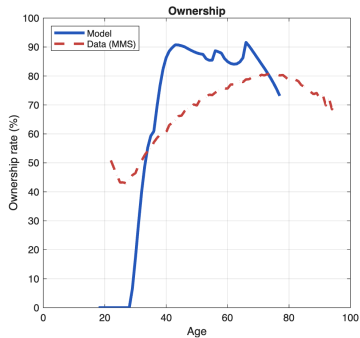
| Param                   | Description              | Value |
|-------------------------|--------------------------|-------|
| $\mathcal{E}_C$         | Center amenity           | 0.30  |
| $\bar{r}_C$             | Center rent anchor       | 0.09  |
| $\beta$                 | Discount factor          | 0.935 |
| $b_0$                   | Entry wealth             | 0.122 |
| $\chi$                  | Owner utility premium    | 1.047 |
| $\psi_{\text{child}}$   | Flow utility of children | 0.067 |
| $\kappa_f$              | Fertility Gumbel scale   | 2.33  |
| $\bar{h}_0$             | Baseline housing need    | 2.52  |
| $\bar{h}_{\text{jump}}$ | Housing jump (1st child) | 0.622 |
| $\bar{h}_n$             | Add'l child housing cost | 0.909 |
| $\bar{c}_n$             | Add'l child cons. cost   | 0.092 |
| $\kappa_\ell$           | Location Gumbel scale    | 1.50  |
| $\mu_{\text{move}}$     | Moving cost              | 5.00  |
| $\theta_0$              | Bequest shifter          | 0.530 |
| $\theta_n$              | Bequest parent premium   | 0.250 |

# Calibration Targets

|                  | Moment                    | Target | Model |
|------------------|---------------------------|--------|-------|
| <i>Fertility</i> | TFR                       | 1.70   | 1.66  |
|                  | Ever childless            | 0.15   | 0.21  |
|                  | Age at first birth        | 26.0   | 34.8  |
|                  | Fertility gradient $P-C$  | 0.133  | 0.168 |
| <i>Housing</i>   | Ownership, ages 30–55     | 0.627  | 0.751 |
|                  | Ownership gradient $P-C$  | 0.170  | 0.185 |
|                  | $H_{01}$ (PSID, rooms)    | 0.664  | 0.323 |
|                  | $H_{12}$ (PSID, rooms)    | 0.566  | 0.299 |
|                  | Parent ownership gap      | 0.110  | 0.184 |
| <i>Saving</i>    | Liq. wealth / income      | 0.60   | 0.460 |
| <i>Location</i>  | Center population share   | 0.450  | 0.454 |
|                  | Rent/room ratio $C/P$     | 1.14   | 1.154 |
|                  | Center share, non-parents | 0.494  | 0.418 |
|                  | Center share, new parents | 0.416  | 0.392 |
|                  | C–P switching rate        | 0.032  | 0.027 |
| <i>Bequest</i>   | Ownership, ages 65–75     | 0.863  | 0.850 |
|                  | Parent–childless own gap  | 0.070  | 0.052 |

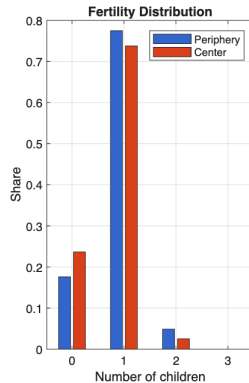
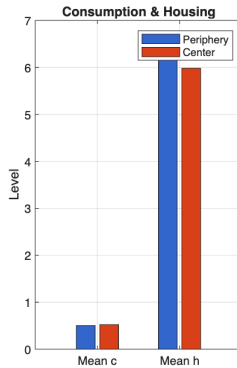
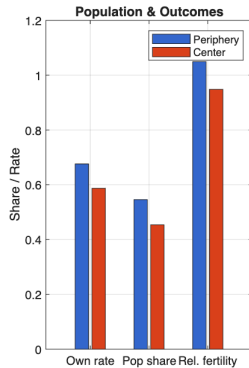
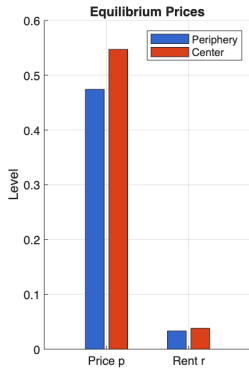
# Lifecycle Fit

Housing Fit

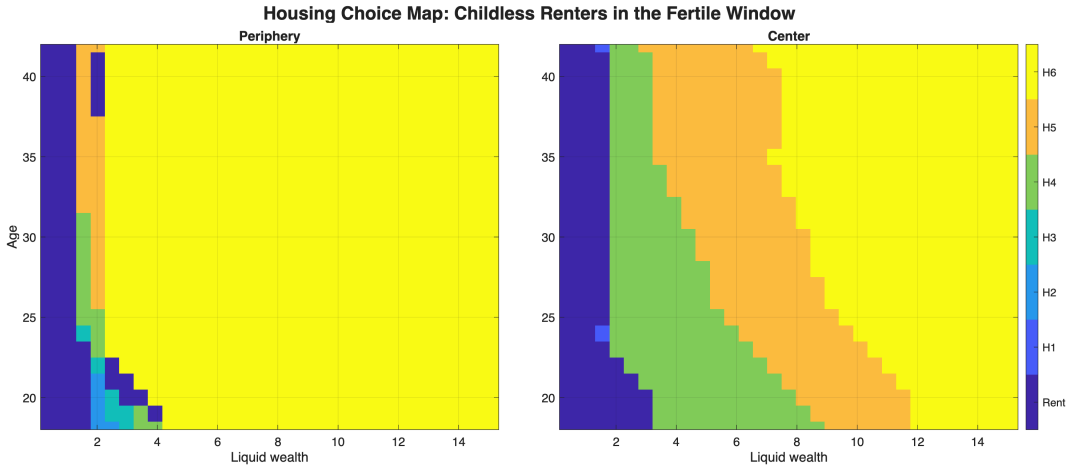


# Equilibrium

## Spatial Equilibrium



# Housing Choice Map in the Fertile Window



# Conclusion

- A model of fertility with realistic housing and lifecycle
- Trying to capture the many implications that children's impact on housing demand has
- Aim is policy:
  - Suppose we want to increase fertility
  - How can housing help? What costs?
- More work is needed:
  - Theory: important to understand the role of constraints and sorting incentives. Can we say something about policy design?
  - Empirically: use new geocoded data to track location and moves
  - Quantification: use the model to evaluate policies that target housing to parents, and understand the channels

**Thank You**

## **Appendix: Theory Details, PSID Details, and Robustness**

## Appendix: Outside-Option Entry Algorithm

1. **Solve normalized benchmark.** Set  $S^* = 1$  and solve for the benchmark composition  $g^*$  and prices  $p^*$ .
2. **Compute per-unit flows.** The benchmark composition implies  $E_0^*$ , the replacement inflow of young adults, and  $B_0^*$ , the mature city-born potential entrant flow.
3. **Choose outside value.** Choose  $\bar{W}^E$  so entry values  $\{W_i^E(p^*)\}_{i \in \{C, P\}}$  imply the target city-entry probability  $q^{E,*}$ .
4. **Set outside-origin flow.** With  $S^* = 1$ ,

$$M = \frac{E_0^*}{q^{E,*}} - B_0^*.$$

5. **Counterfactuals.** Hold  $(M, \bar{W}^E, \kappa_E)$  fixed and let  $q^E(p)$  and  $S(p)$  respond.

## Appendix: Aggregate vs. Location Stationarity

- The scalar equation pins down total city scale:

$$SE_0(p) = q^E(p) [M + SB_0(p)].$$

- Location-specific entry masses use the unconditional option probabilities:

$$E_i(G; p) = [M + SB_0(p)] \pi_i^E(p), \quad i \in \{C, P\}.$$

- Thus  $q^E(p)$  answers *how many* potential entrants enter the modeled metro, while  $\pi_C^E(p)$  and  $\pi_P^E(p)$  answer *where* they enter inside the metro.
- The distribution recursion uses the vector  $(E_C, E_P)$ , so aggregate stationarity alone is not the whole equilibrium object.

# Calibration Strategy

## Externally set

- Lifecycle structure, taxes, interest rate
- Collateral ( $\phi = 0.80$ ), transaction cost ( $\psi = 0.06$ )
- Housing ladder and supply elasticities
- Five owner sizes at  $\{5.2, 5.85, 6.55, 7.55, 9.0\}$  rooms; rental cap at 4.8 rooms.
- Bequests parameterized by  $(\theta_0, \theta_n)$  with fixed  $\theta_1 = 0.01$ .

## Internally calibrated (SMM)

- 15 parameters: 2 inverted (amenity, rent anchor) + 13 searched
- Fertility, tenure, mobility, saving, bequests
- 17 target moments from ACS and PSID

## Appendix: Setup — Utility, Demands, Indirect Utility

[back](#)

Common preliminaries for the derivations that follow.

**Utility** (CRRA over Cobb–Douglas Stone–Geary):

$$u(c, h; m) = \frac{[(c - \bar{c}(m))^\alpha (h - \bar{h}(m))^{1-\alpha}]^{1-\sigma}}{1 - \sigma}.$$

**Interior demands** ( $\sigma$ -invariant). With  $M$  household resources, rent  $r_i$  in location  $i$ , and  $\tilde{M}_i(\bar{h}) = M - \bar{c} - r_i \bar{h}$ :

$$c^* - \bar{c} = \alpha \tilde{M}_i, \quad h^* - \bar{h} = (1 - \alpha) \tilde{M}_i / r_i.$$

**Indirect utility with amenity**  $\mathcal{E}_i$ :

$$\mathcal{V}_i(\bar{h}; M) = \frac{Z^*(\bar{h}; M, r_i)^{1-\sigma}}{1 - \sigma} + \mathcal{E}_i, \quad Z^* = \alpha^\alpha (1 - \alpha)^{1-\alpha} \tilde{M}_i r_i^{-(1-\alpha)}.$$

## Appendix: Sorting Derivation [back](#)

For  $r_C > r_P$ ,  $\mathcal{E}_C > \mathcal{E}_P$ , define  $\Delta\mathcal{V}(\bar{h}; M) = \mathcal{V}_C(\bar{h}; M) - \mathcal{V}_P(\bar{h}; M)$ .

**Derivative.** With  $\gamma \equiv \alpha + \sigma(1 - \alpha) > 0$  and  $\kappa \equiv \alpha^\alpha(1 - \alpha)^{1-\alpha}$ ,

$$\frac{d\Delta\mathcal{V}}{d\bar{h}} = -\kappa^{1-\sigma} \left[ \frac{r_C^\gamma}{\tilde{M}_C^\sigma} - \frac{r_P^\gamma}{\tilde{M}_P^\sigma} \right] < 0.$$

**Sign.**  $r_C > r_P$  implies  $r_C^\gamma > r_P^\gamma$ ;  $\tilde{M}_C < \tilde{M}_P$  implies  $\tilde{M}_C^{-\sigma} > \tilde{M}_P^{-\sigma}$ . Both factors favour C, so the bracket is strictly positive and the derivative is strictly negative.

**Threshold.** Since  $\Delta\mathcal{V}$  is continuous and strictly decreasing in  $\bar{h}$ , the equation  $\Delta\mathcal{V}(\hat{h}; M) = 0$  has at most one solution. Existence on a non-degenerate interval follows from  $\Delta\mathcal{V}(0; M) > 0$  (amenity-positive) and  $\Delta\mathcal{V}(\bar{h}; M) \rightarrow -\infty$  as  $\bar{h} \uparrow (M - \bar{c})/r_C$  (infeasibility in C). Unique threshold  $\hat{h}(M)$ .

**Parent-childless ordering.**  $\bar{h}(1) > \bar{h}(0)$  and  $\Delta\mathcal{V}$  strictly decreasing  $\Rightarrow$  parents weakly prefer the periphery, strictly when  $\bar{h}(0) < \hat{h}(M) < \bar{h}(1)$ .

## Appendix: Elasticity-Attenuation Derivation

[back](#)

**Simplified demand** (interior renter at rent  $r$ ; assume  $\bar{c}(0) = \bar{c}(1) = \bar{c}$  so that only the housing requirement varies with the number of children):

$$h^*(n) = \alpha \bar{h}(n) + (1 - \alpha) \frac{M - \bar{c}}{r}.$$

**Elasticities.**

$$e_M(n) = \frac{\partial \log h^*}{\partial \log M} = \frac{(1 - \alpha)(M - \bar{c})}{r h^*(n)}, \quad e_r(n) = \frac{\partial \log h^*}{\partial \log r} = - \frac{(1 - \alpha)(M - \bar{c})}{r h^*(n)}.$$

**Attenuation.**  $h^*(n)$  is strictly increasing in  $\bar{h}(n)$  (coefficient  $\alpha > 0$ ); the numerator is independent of  $\bar{h}(n)$ . Hence both  $e_M(n)$  and  $|e_r(n)|$  are strictly decreasing in  $\bar{h}(n)$ . Since  $\bar{h}(1) > \bar{h}(0)$ :

$$e_M(1) < e_M(0), \quad |e_r(1)| < |e_r(0)|.$$

**Limits.** As  $\bar{h} \rightarrow 0$ , both elasticities  $\rightarrow \mp 1$  (Cobb–Douglas benchmark). As  $\bar{h} \rightarrow (M - \bar{c})/r$  (feasibility boundary), both  $\rightarrow 0$  (pure subsistence).

**Identification.** The gap  $e_M(0) - e_M(1) > 0$  pins down  $\Delta \bar{h} = \bar{h}(1) - \bar{h}(0)$  given  $\alpha$ ,  $M - \bar{c}$ , and  $r$ .

## Appendix: Cross-Metro Sorting-Wedge Derivation

[back](#)

Indirect utility on an interior renter branch in location  $i$  with  $n$  children ( $\sigma = 1$  for transparency; signs go through for any  $\sigma > 0$ ):

$$\mathcal{V}_i(n; M) = \log \tilde{M}_i^n - (1 - \alpha) \log r_i + \mathcal{E}_i, \quad \tilde{M}_i^n = M - \bar{c}(n) - r_i \bar{h}(n).$$

**Sorting wedge.** Define  $\Delta \mathcal{V}(n; M) = \mathcal{V}_C(n; M) - \mathcal{V}_P(n; M)$  and  $\mathcal{W}(M) \equiv \Delta \mathcal{V}(0; M) - \Delta \mathcal{V}(1; M)$ :

$$\mathcal{W}(M) = \log \frac{\tilde{M}_C^0 \tilde{M}_P^1}{\tilde{M}_C^1 \tilde{M}_P^0}.$$

**Comparative statics in rents.** Using  $\partial \tilde{M}_i^n / \partial r_i = -\bar{h}(n)$ ,

$$\frac{\partial \mathcal{W}}{\partial r_C} = \frac{\delta_h(M - \bar{c}(0)) + \bar{h}(0) \delta_c}{\tilde{M}_C^1 \tilde{M}_C^0} > 0, \quad \frac{\partial \mathcal{W}}{\partial r_P} = -\frac{\delta_h(M - \bar{c}(0)) + \bar{h}(0) \delta_c}{\tilde{M}_P^1 \tilde{M}_P^0} < 0.$$

Numerators positive by interior feasibility ( $M > \bar{c}(0)$ ) plus  $\delta_c, \delta_h, \bar{h}(0) > 0$ ; denominators positive on the interior renter branch.

**Reading.** Raising the centre rent (or lowering the periphery rent) widens the parent-childless sorting wedge: the cross-metro elasticity of sorting strength to the local rent gap is strictly positive. Same algebraic structure carries through for capped renters (replace  $\tilde{M}_i^n$  with  $\check{M}_i^n = M - r_i \bar{h}_R - \bar{c}(n)$ ).

## Appendix: Constrained-Squeeze CEV Derivation

[back](#)

CEV solves  $u(c, h, 0) = u(c + \text{CEV}, h, 1)$ : housing quantity is held fixed at  $h$ , and the post-child surplus shrinks via  $\bar{h}(n)$  from  $s^0 = h - \bar{h}(0)$  to  $s^0 - \Delta\bar{h} = h - \bar{h}(1)$ . Under Stone–Geary  $u = [(c - \bar{c}(n))^\alpha (h - \bar{h}(n))^{1-\alpha}]^{1-\sigma} / (1 - \sigma)$  and  $\bar{c}(0) = \bar{c}(1) = \bar{c}$ , equating utilities:

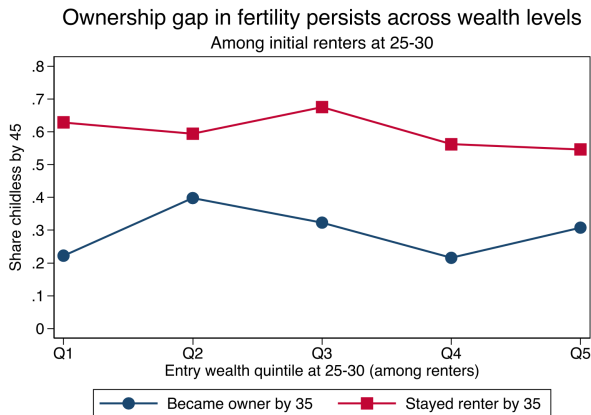
$$(c - \bar{c})^\alpha (s^0)^{1-\alpha} = (c + \text{CEV} - \bar{c})^\alpha (s^0 - \Delta\bar{h})^{1-\alpha}.$$

Rearrange:

$$\frac{c + \text{CEV} - \bar{c}}{c - \bar{c}} = \left( \frac{s^0}{s^0 - \Delta\bar{h}} \right)^{(1-\alpha)/\alpha} \implies \text{CEV} = (c - \bar{c}) \left[ \left( \frac{s^0}{s^0 - \Delta\bar{h}} \right)^{(1-\alpha)/\alpha} - 1 \right].$$

# Ownership Is the Binding Constraint on Fertility

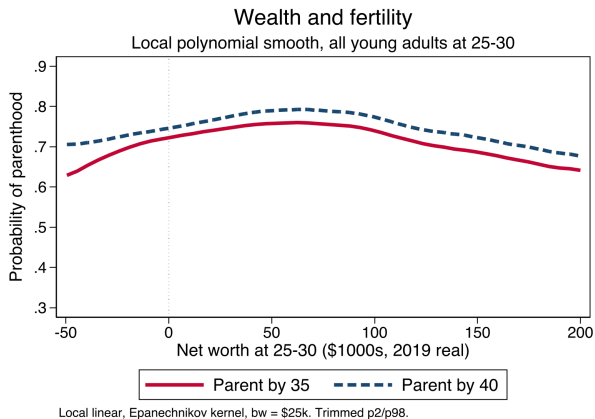
Aggregate Gap



Sample: PSID renters at ages 25–30, pre-first-birth, followed to age 45 ( $N = 1,127$ ).

- At **every** wealth quintile, renters who transition to ownership by 35 are 20–35 pp less likely to remain childless by 45.
- The gap does not shrink with wealth: even wealthy renters who do not buy remain  $\sim 55\%$  childless.
- Mediation regression: wealth has **zero** direct effect on fertility ( $p = 0.92$ ): the entire effect runs<sup>9/30</sup>

# Wealth and Fertility: Two Channels

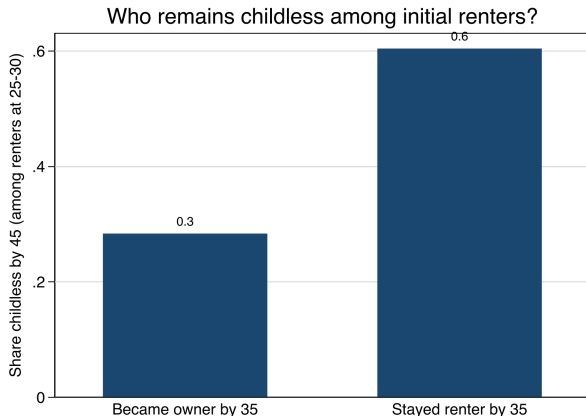


The unconditional wealth–fertility relationship is **hump-shaped**:

- **Rising part** ( $NW < \$80k$ ):  
more wealth  $\Rightarrow$  can clear down payment  $\Rightarrow$  access family-sized housing  $\Rightarrow$  have children. The *resource/housing channel*.
- **Falling part** ( $NW > \$80k$ ):  
wealthier young adults are disproportionately high human capital and career-invested. The *opportunity cost channel*.

The non-monotonicity is why we need

# Aggregate Ownership Transition Gap

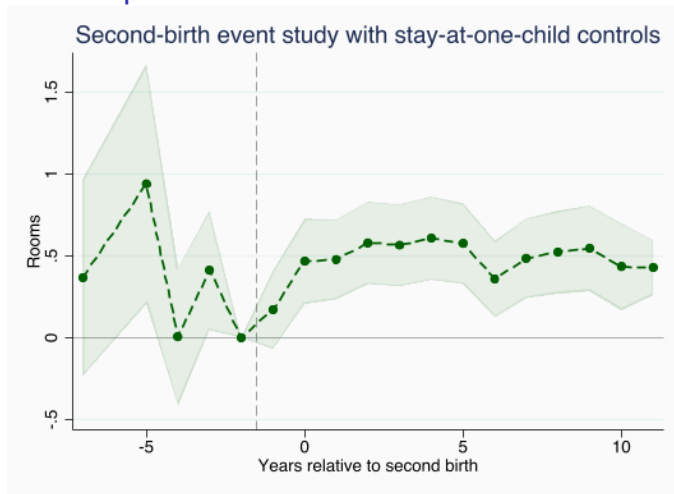
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Among renters at 25–30 who have not yet had a first child:

- Those who transition to ownership by 35: **28%** childless at 45.
- Those who stay renters through 35: **60%** childless at 45.
- A **32 pp** extensive-margin gap—larger than any within-tenure wealth effect.

Tenure status averaged over ages 25–30 (renter) and 31–35 (transition outcome). Childlessness measured by 45 using reported age at first birth.

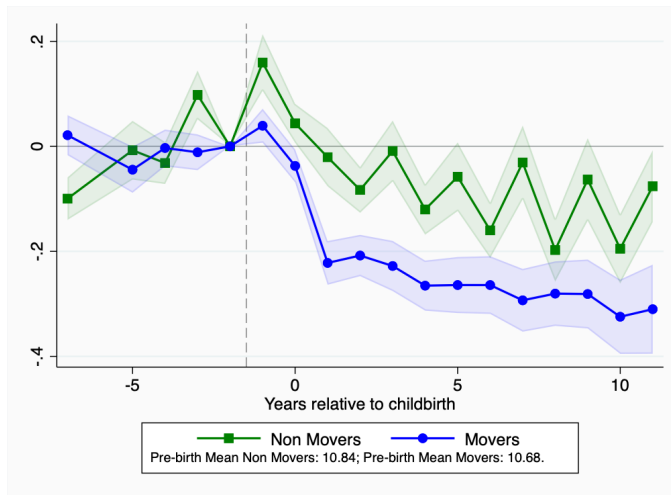
## PSID: Housing Also Expands After Second Birth

[Back](#)

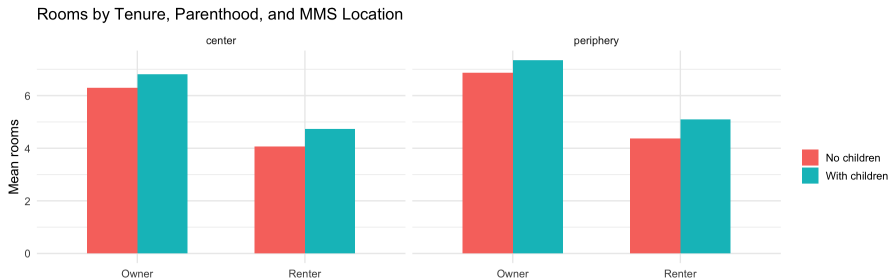
Baseline second-birth event study with stay-at-one-child controls: the +3 coefficient is about **0.57 rooms**.

# PSID: Income Trajectories, Movers vs Non-Movers

No IDFE



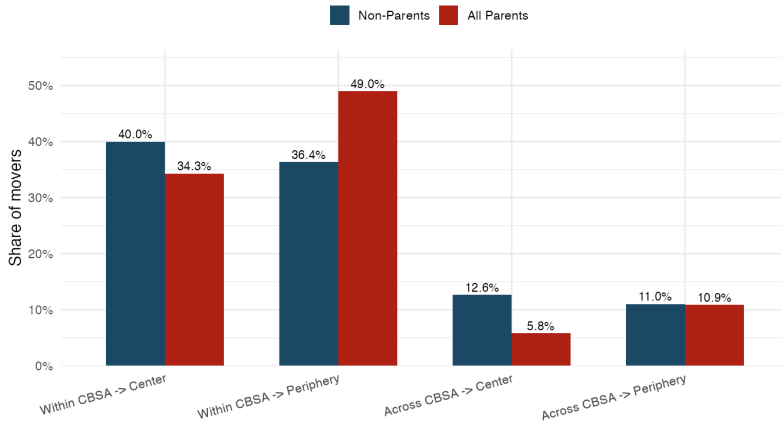
# Defining Center and Periphery Within Large Metros

[Back](#)

Sample: U.S. metros with 2000 population  $> 1M$  (51 metros). Core = tracts containing the closest 30% of metro population to the CBD. A PUMA is classified as center if at least 10% of its population lives in core tracts. Center pop share: 45.0%.

# Most Adjustment Is Within Metros

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- Most adjustment happens **within** metros rather than across metros.

# Mover Regressions: Parent vs. Non-Parent

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| Outcome (pp)                     | (1)<br>New Parent   | (2)<br>Any Parent   |
|----------------------------------|---------------------|---------------------|
| Across-CBSA move                 | −4.56***<br>(0.78)  | −5.17***<br>(0.54)  |
| Center destination   within-CBSA | −11.21***<br>(0.66) | −11.44***<br>(0.53) |
| Within CBSA → Center             | −6.77***<br>(0.77)  | −6.79***<br>(0.65)  |
| Within CBSA → Periphery          | 11.33***<br>(0.49)  | 11.95***<br>(0.30)  |
| Center dest.   center origin     | −9.69***<br>(0.81)  | −8.91***<br>(0.53)  |
| Center dest.   periphery origin  | −10.71***<br>(0.51) | −11.32***<br>(0.47) |
| Age and year FE                  | Yes                 | Yes                 |
| Sex and race controls            | Yes                 | Yes                 |

*Notes.* OLS on ACS 2012–2023 within-metro movers, ages 22–45, weighted by *perwt*. Coefficients in percentage points. Column (1) uses a *new-parent* indicator (eldest child < 4); column (2) uses an *any-parent* indicator. Standard errors in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

# Cross-Metro Sorting Regression

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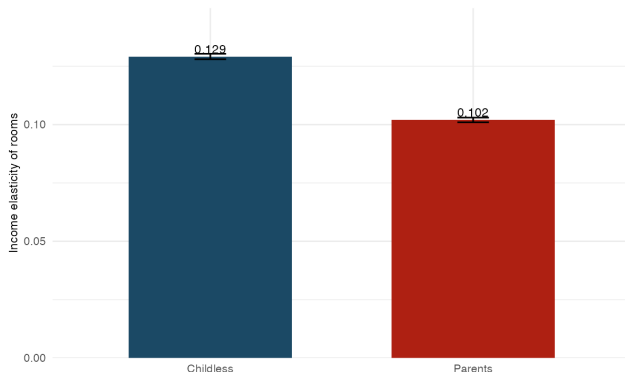
| Dependent gap (pp)                          | Slope on log rent gap | Metros |
|---------------------------------------------|-----------------------|--------|
| Center-share gap                            | 6.23<br>(5.19)        | 42     |
| Center-destination gap   within-CBSA movers | 23.53***<br>(6.18)    | 42     |

*Notes.* Unit of observation is the metro. Each dependent variable is the non-parent minus new-parent gap in percentage points. Regressor is the local log center-periphery rent-per-room gap. Each regression is weighted by its corresponding metro sample size. \*\*\*  $p < 0.01$ .

# Housing Demand Elasticity Among Renters

[Back](#)[Regression](#)

*Model: parents less income-elastic in housing ( $\bar{h}(1) > \bar{h}(0)$ ).*



Mean rooms among renters: 4.20 for childless households, 4.92 for parents.

- Parent renters start from a larger space baseline, but rooms rise less with income: elasticity 0.102 for parents versus 0.129 for childless renters.

# Renter Housing-Demand Elasticity

[Back](#)

| Outcome               | Childless slope     | Parent interaction   | Parent slope        |
|-----------------------|---------------------|----------------------|---------------------|
| Log rooms             | 0.129***<br>(0.001) | −0.027***<br>(0.001) | 0.102***<br>(0.001) |
| Metro FE              | Yes                 | Yes                  | Yes                 |
| Age and year FE       | Yes                 | Yes                  | Yes                 |
| Sex and race controls | Yes                 | Yes                  | Yes                 |
| Observations          |                     | 3,024,081            |                     |

*Notes.* OLS on ACS renters, ages 22–45, 2012–2023, weighted by perwt. Specification is  $\log(\text{rooms})$  on  $\log(\text{income})$ , parent status, and the interaction, with metro, age, and year fixed effects plus sex and race controls. Standard errors in parentheses. \*\*\*  $p < 0.01$ .

# Potential Reasons to Move in the PSID Questionnaire

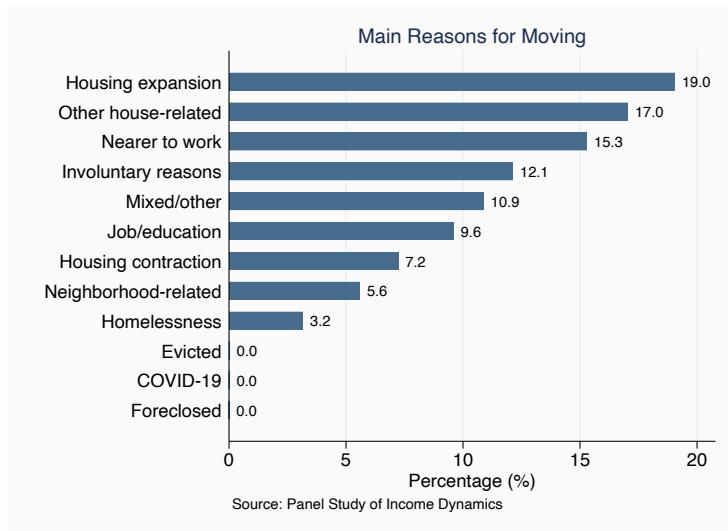
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Movers are asked: “What is the main reason you moved?”. Categories:

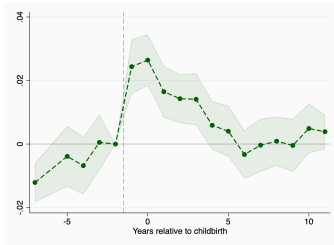
- **Purposive productive:** take another job; transfer; stopped going to school.
- **Nearer to work.**
- **Purposive consumptive — expansion of housing:** more space; higher rent; better place.
- **Purposive consumptive — contraction of housing:** less space; less rent.
- **Purposive consumptive — other house-related:** get own home; got married; physical condition of prior unit.
- **Purposive consumptive — neighborhood:** better neighborhood; schooling; closer to friends/relatives.
- **Involuntary:** housing unit coming down; armed services; health; divorce; forced retirement.
- **Ambiguous or mixed:** save money, neighbors moved away, retiring.
- Became homeless or no longer homeless; COVID-19; evicted / foreclosed.

# Distribution of Reasons to Move Among Non-Parents

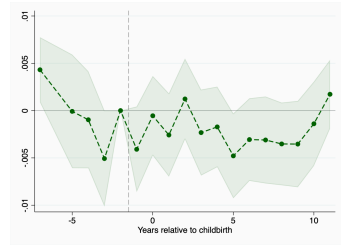
This tabulates values only for non-parents (both “never” and “not yet”).



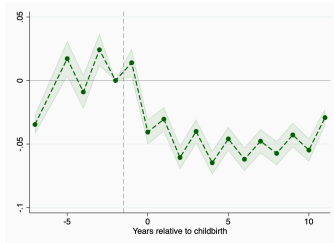
# Event Studies: Housing and Location Reasons

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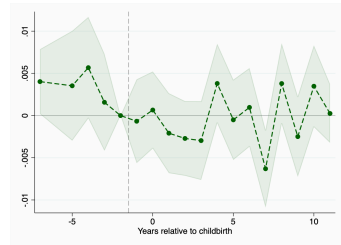
Expansion of housing



Contraction of housing

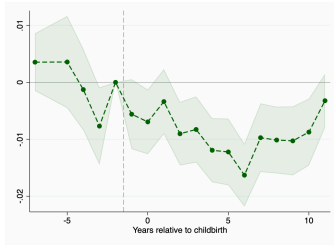


Other house-related

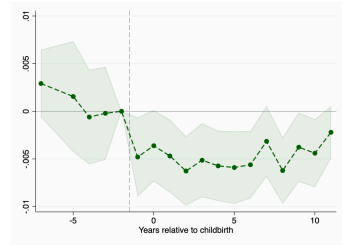


Neighborhood

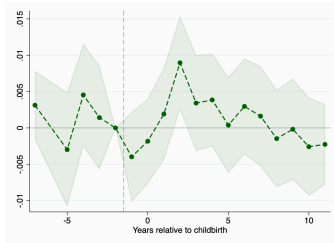
# Event Studies: Work and Involuntary Reasons



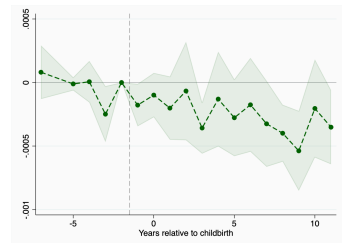
Productive reasons



To get nearer to work

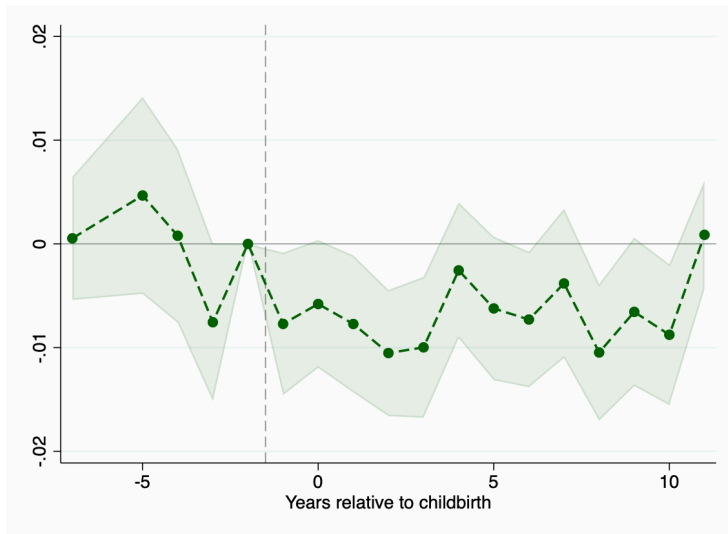


Involuntary reasons



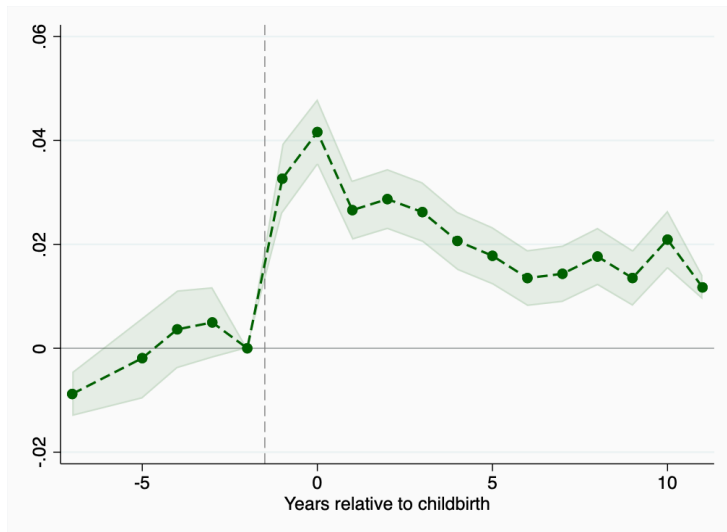
Was evicted

## Event Studies: Other Reasons

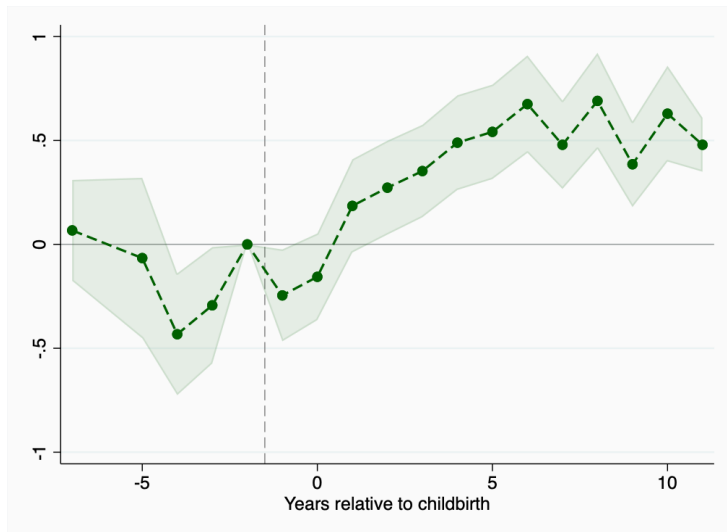


Mixed or other reasons

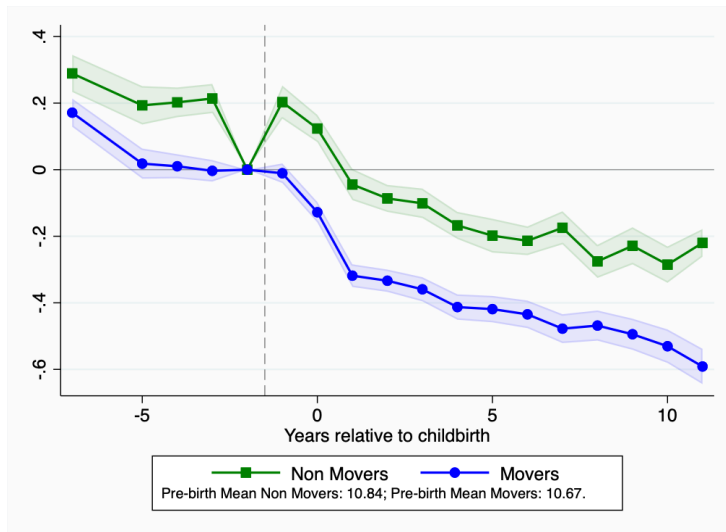
## No Individual FE: Reasons to Move (Expansion of Housing)

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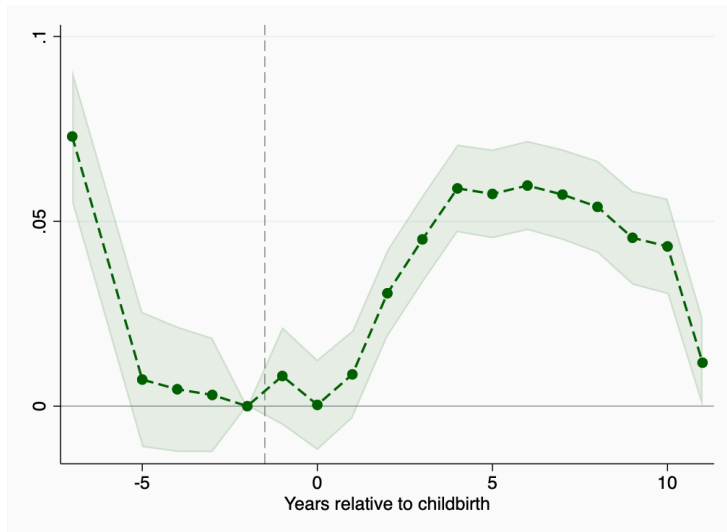
## No Individual FE: Number of Rooms Around First Birth

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# No Individual FE: Family Income Around First Birth

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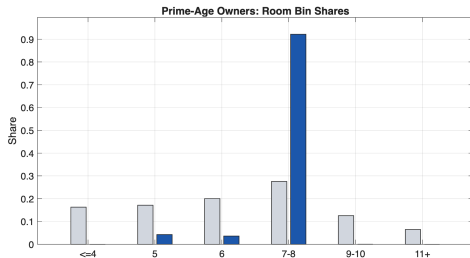
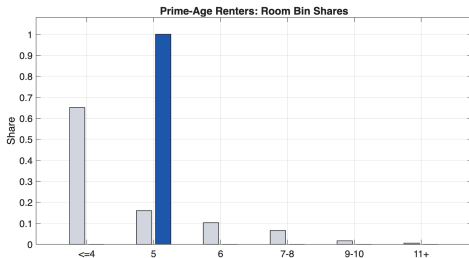
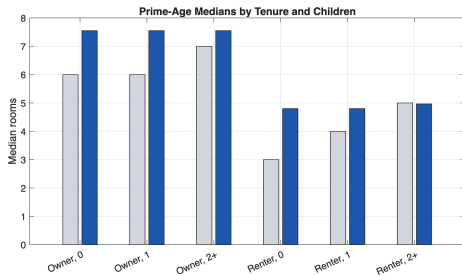
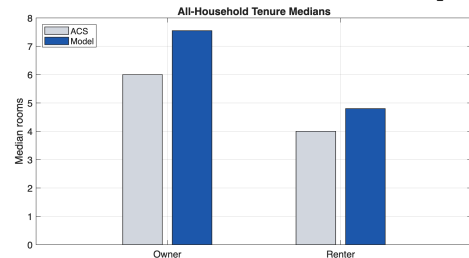
## No Individual FE: Homeownership Around First Birth

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# Housing Size Fit: Model vs Data

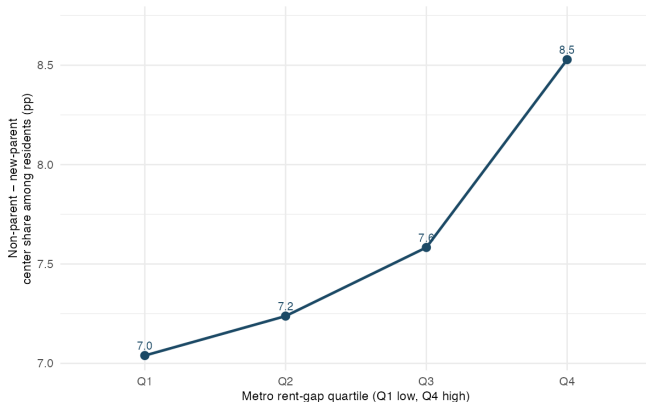
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Housing Size Fit: Model vs Data



## Sorting Strength: Residents (Stock) [Back](#)

*Model: parent share falls with local rent;  $\partial \mathcal{W} / \partial r_C > 0$ .*



Y-axis: non-parent minus new-parent center share among all residents (pp). Base: 49.4% for non-parents, 41.6% for new parents.

- Stock-based sorting gap also rises with the rent gap, but the gradient is flatter than among movers.